

Contact Information	Halicioğlu Data Science Institute University of California, San Diego 3234 Matthews Lane La Jolla, CA 92093	☎ 203-305-9456 ✉ tianhaowang@ucsd.edu 📄 https://tiiao.github.io 🔗 Google Scholar
Employment	Assistant Professor, Halicioğlu Data Science Institute, University of California, San Diego Research Assistant Professor, Toyota Technological Institute at Chicago	2025 – present 2024 – 2025
Education	Ph.D. in Statistics and Data Science, Yale University • Advisor: Prof. Zhou Fan M.A. in Statistics (en route), Yale University B.S. in Mathematics, University of Science and Technology of China (USTC) • B.E. in Computer Science (dual)	2024 2021 2014 – 2018
Publications and Preprints	(*: equal contribution) 1. Shuo Xie*, Tianhao Wang*, Beining Wu, and Zhiyuan Li. “A Tale of Two Geometries: Adaptive Optimizers and Non-Euclidean Descent”. <i>International Conference on Learning Representations (ICLR)</i> , 2026. Available at arXiv: 2511.20584. 2. Siyu Chen, Heejune Sheen, Xuyuan Xiong, Tianhao Wang, and Zhuoran Yang. “Taming Polysemanticity in LLMs: Provable Feature Recovery via Sparse Autoencoders”. <i>International Conference on Learning Representations (ICLR)</i> , 2026. Available at arXiv: 2506.14002. 3. Xinhe Wang, Jin Huang, Xingjian Zhang, Tianhao Wang, Jiaqi W. Ma. “Your Reasoning Benchmark May Not Test Reasoning: Revealing Perception Bottleneck in Abstract Reasoning Benchmarks”. Available at arXiv: 2512.21329. 4. Mohamad Amin Mohamadi, Tianhao Wang, and Zhiyuan Li. “Honesty over Accuracy: Trustworthy Language Models through Reinforced Hesitation”. Available at arXiv: 2511.11500. 5. Xingjian Zhang, Tianhong Gao, Suliang Jin, Tianhao Wang, Teng Ye, Eytan Adar, and Qiaozhu Mei. “Through the Judge’s Eyes: Inferred Thinking Traces Improve Reliability of LLM Raters”. Available at arXiv: 2510.25860. 6. Robin Yadav, Shuo Xie, Tianhao Wang, and Zhiyuan Li. “Provable Benefit of Sign Descent: A Minimal Model Under Heavy-Tail Class Imbalance”. <i>NeurIPS 2025 Workshop on Optimization for Machine Learning (OPT 2025)</i> . (Oral) . 7. Max Lovig, Tianhao Wang, and Zhou Fan. “On Universality of Non-Separable Approximate Message Passing Algorithms”. Available at arXiv: 2506.23010. 8. Shuo Xie, Tianhao Wang, Sashank Reddi, Sanjiv Kumar, and Zhiyuan Li. “Structured Preconditioners in Adaptive Optimization: A Unified Analysis”. <i>International Conference on Machine Learning (ICML)</i> , 2025. Available at arXiv: 2503.10537. 9. Siyu Chen*, Beining Wu*, Miao Lu, Zhuoran Yang, and Tianhao Wang. “Can Neural Networks Achieve Optimal Computational-statistical Tradeoff? An Analysis on Single-Index Model”. <i>International Conference on Learning Representations (ICLR)</i> , 2025 (Oral) . 10. Angeliki Giannou, Liu Yang, Tianhao Wang, Dimitris Papailiopoulos, and Jason D. Lee. “How well can Transformers emulate in-context Newton’s method?” <i>International Conference on Artificial Intelligence and Statistics (AISTATS)</i> , 2025. Available at arXiv:2403.03183. 11. Siyu Chen*, Heejune Sheen*, Tianhao Wang, and Zhuoran Yang. “Unveiling Induction Heads: Provable Training Dynamics and Feature Learning in Transformers”. <i>Annual Conference on Neural Information Processing Systems (NeurIPS)</i> , 2024. Available at arXiv:2409.10559. 12. Xinyi Zhong*, Tianhao Wang*, and Zhou Fan. “Approximate Message Passing for orthogonally invariant ensembles: Multivariate non-linearities and spectral initialization”. <i>Information and Inference</i> , 2024. Available at arXiv:2110.02318.	

13. Tianhao Wang, Xinyi Zhong, and Zhou Fan. “Universality of Approximate Message Passing algorithms and tensor networks”. *Annals of Applied Probability*, 2024. Available at arXiv:2206.13037.
14. Heejune Sheen, Siyu Chen, Tianhao Wang, and Harrison H. Zhou. “Implicit Regularization of Gradient Flow on One-Layer Softmax Attention”. Available at arXiv:2403.08699.
15. Siyu Chen, Heejune Sheen, Tianhao Wang, and Zhuoran Yang. “Training dynamics of multi-head softmax attention for in-context learning: emergence, convergence, and optimality”. *Conference on Learning Theory (COLT)*, 2024. Available at arXiv:2402.19442.
16. Zhou Fan, Roy R. Lederman, Yi Sun, Tianhao Wang, Sheng Xu. “Maximum likelihood for high-noise group orbit estimation and single-particle cryo-EM”. *Annals of Statistics*, 2024. Available at arXiv:2107.01305.
17. Runzhe Wang, Sadhika Malladi, Tianhao Wang, Kaifeng Lyu, and Zhiyuan Li. “The Marginal Value of Momentum for Small Learning Rate SGD”. *International Conference on Learning Representations (ICLR)*, 2024. Available at arXiv:2307.15196.
18. Zhou Fan, Yi Sun, Tianhao Wang, Yihong Wu. “Likelihood landscape and maximum likelihood estimation for the discrete orbit recovery model”. *Communications on Pure and Applied Mathematics*, 2023. Available at arXiv:2004.00041.
19. Ruitu Xu, Yifei Min, and Tianhao Wang. “Noise-adaptive Thompson sampling for linear contextual bandits”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
20. Yifei Min, Jiafan He, Tianhao Wang, Quanquan Gu. “Cooperative multi-agent reinforcement learning: Asynchronous communication and linear function approximation”. *International Conference on Machine Learning (ICML)*, 2023. Available at arXiv:2305.06446.
21. Ruitu Xu, Yifei Min, Tianhao Wang, Michael I. Jordan, Zhaoran Wang, Zhuoran Yang. “Finding regularized competitive equilibria of heterogeneous agent macroeconomic models via reinforcement learning”. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023. Available at arXiv:2303.04833.
22. Zhiyuan Li, Tianhao Wang, Dingli Yu. “Fast mixing of stochastic gradient descent with normalization and weight decay”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2022.
23. Zhiyuan Li*, Tianhao Wang*, Jason D. Lee, Sanjeev Arora. “Implicit bias of gradient descent on reparametrized models: on equivalence to mirror descent”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2022. Available at arXiv:2207.04036.
24. Jiafan He*, Tianhao Wang*, Yifei Min*, Quanquan Gu. “A simple and provably efficient algorithm for asynchronous federated linear bandits”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2022. Available at arXiv:2207.03106.
25. Yifei Min, Tianhao Wang, Ruitu Xu, Zhaoran Wang, Michael I. Jordan, Zhuoran Yang. “Learn to match with no regret: Reinforcement learning in Markov matching market”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2022 (**Oral**). Available at arXiv:2203.03684.
26. Yifei Min, Jiafan He, Tianhao Wang, Quanquan Gu. “Learning stochastic shortest path with linear function approximation”. *International Conference on Machine Learning (ICML)*, 2022. Available at arXiv:2110.12727.
27. Zhiyuan Li, Tianhao Wang, Sanjeev Arora. “What happens after SGD reaches zero loss? – A mathematical framework”. *International Conference on Learning Representations (ICLR)*, 2022 (**Spotlight**). Available at arXiv:2110.06914.
28. Pamela L Valentino, Tianhao Wang, Veronika Shabanova, Vicky Lee Ng, John C Bucuvalas, Amy G Feldman, Regino P Gonzalez-Peralta, Nitika Arora Gupta, Tamir A Miloh, Saeed Mohammad, Erika Pace, Shikha S Sundaram, Nada A Yazigi, Kyle Soltys, Society of Pediatric Liver Transplantation (SPLIT). “North American biliary stricture management strategies in children post liver transplant: multicenter analysis from the SPLIT Registry”. *Liver Transplantation*, 2021.
29. Yifei Min*, Tianhao Wang*, Dongruo Zhou, Quanquan Gu. “Variance-aware off-policy evaluation with linear function approximation”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2021. Available at arXiv:2106.11960.
30. Tianhao Wang*, Dongruo Zhou*, Quanquan Gu. “Provably efficient reinforcement learning with linear function approximation under adaptivity constraints”. *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2021. Available at arXiv:2101.02195.

31. Pan Xu*, Tianhao Wang*, Quanquan Gu. “Continuous and discrete-time accelerated stochastic mirror descent for strongly convex functions”. *International Conference on Machine Learning (ICML)*, 2018.
32. Pan Xu*, Tianhao Wang*, Quanquan Gu. “Accelerated stochastic mirror descent: From continuous-time dynamics to discrete-time algorithms”. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2018.

Honors and Awards

- Leonard F. Savage Prize for best written work Yale, 2024
- Student Poster Competition Award Rutgers University, 2023
 - At Conference on Recent Advances in Statistics and Data Science
- NeurIPS 2022 top reviewer NeurIPS, 2022
- Wedworth W. Clarke Fellowship Yale, 2021

Invited Talks and Seminars

- International Conference on Continuous Optimization, Los Angeles 07/2025
- Midwest Machine Learning Symposium, Chicago 06/2025
- International Indian Statistical Association Conference, Lincoln 06/2025
- Midwest Optimization & Statistical Learning Conference, Chicago 05/2025
- Flatiron Institute, CCM ML seminar 04/2025
- Research at TTIC 11/2024
- FAI seminar (online) 08/2024
- Joint Statistical Meetings, Portland 08/2024
- University of Notre Dame, ACMS Colloquium 02/2024
- UIUC, Statistics Seminar 02/2024
- Columbia University, Statistics Seminar 01/2024
- UCSD, Halicioğlu Data Science Institute, Special Seminar Series 01/2024
- UC Davis, Statistics Seminar 01/2024
- UCSD, Department of Mathematics Colloquium 12/2023
- INFORMS Annual Meeting, Phoenix 10/2023

Teaching

- Instructor, University of California, San Diego
- DSC 242: High-dimensional Probability and Statistics Fall 2025
- Teaching assistant, Yale University
- High-Dimensional Phenomena in Statistics and Learning Spring 2023
 - Intermediate Machine Learning Spring 2022
 - Statistical Inference Fall 2020, Fall 2021
 - Information Theory Spring 2021
 - Probability and Statistics Fall 2019
 - Stochastic Processes Spring 2019, Spring 2020

**Professional
Service**

Organizing committee for

- Midwest Optimization & Statistical Learning Conference 2025

Conference area chair for

- NeurIPS 2025
- ICLR 2025, 2026

Journal reviewer for

- Annals of Statistics
- IEEE Transactions on Information Theory
- Journal of the American Statistical Association
- Journal of Machine Learning Research
- Journal of the Royal Statistical Society: Series B
- Random Matrices: Theory and Applications
- SIAM Journal on Mathematics of Data Science
- Statistical Science

Conference reviewer for

- AAAI 2025
- AISTATS 2022, 2023, 2024
- COLT 2025
- ICLR 2023, 2024, 2025
- ICML 2022, 2023, 2024, 2025, 2026
- IEEE ITW 2023
- NeurIPS 2022, 2023, 2024
- Reinforcement Learning Conference 2024, 2025